Table 10: Critical layers per condition (>20% accuracy drop on knockout).

Model	Set A MHSA	Set B MHSA	Set A FFN	Set B FFN
Llama-1B inst	1	12	4	14
Llama-1B base	2	7	3	10
Llama-8B inst	0	1	4	4
Llama-8B base	0	0	1	2
Gemma-9B inst	0	0	0	0
Gemma-9B base	0	0	0	2

4, and 9B models have 0–2. Third, at larger scale, the remaining critical layers shift from MHSA to FFN—8B and 9B models have more FFN critical layers than MHSA, suggesting that emotion processing shifts from attention-mediated to feedforward-mediated as capacity increases. Random ablation (replacing layer output with noise rather than zeroing) yields the same pattern with uniformly higher counts, confirming that these results are not artifacts of the ablation method.

The scale shift. Larger models show a fundamentally different architecture. In Llama-8B and Gemma-9B, no single-layer knockout is catastrophic for either stimulus type. The emotion processing pathway is distributed, with graceful degradation rather than single points of failure.

This shift from concentrated to distributed processing parallels a structural principle in neuroscience: smaller, less complex neural systems concentrate critical functions in narrow circuits and are vulnerable to focal disruption, while larger systems develop redundant pathways where damage to any single component is compensated by others. We emphasize that this is a structural analogy, not a neuroanatomical claim. But it suggests that scale does not simply add more of the same processing—it reorganizes *how* emotion processing is architecturally distributed.

6.3 Attention Patterns: The Dissociation Is Not Attention-Mediated

Analysis of attention weights at the extraction position reveals a consistent pattern across all models and both stimulus sets. The model's attention is dominated by:

1. `<|begin_of_text|>`—the structural anchor (highest attention weight)
2. Punctuation tokens (`:`, `.`)—structural markers
3. `"joy"` and `"sadness"`—the few-shot example labels
4. `"emotions"`—from the prompt frame

Critically, this pattern is *identical for Set A and Set B*. The model does not attend more to emotion keywords in keyword-rich stimuli, nor does it attend to specific situational cues in keyword-free stimuli. Instead, it attends to the few-shot label space in both conditions.

This is a useful negative finding. Attention patterns do not differentiate between keyword-rich and keyword-free processing, indicating that the dissociation between affect reception and

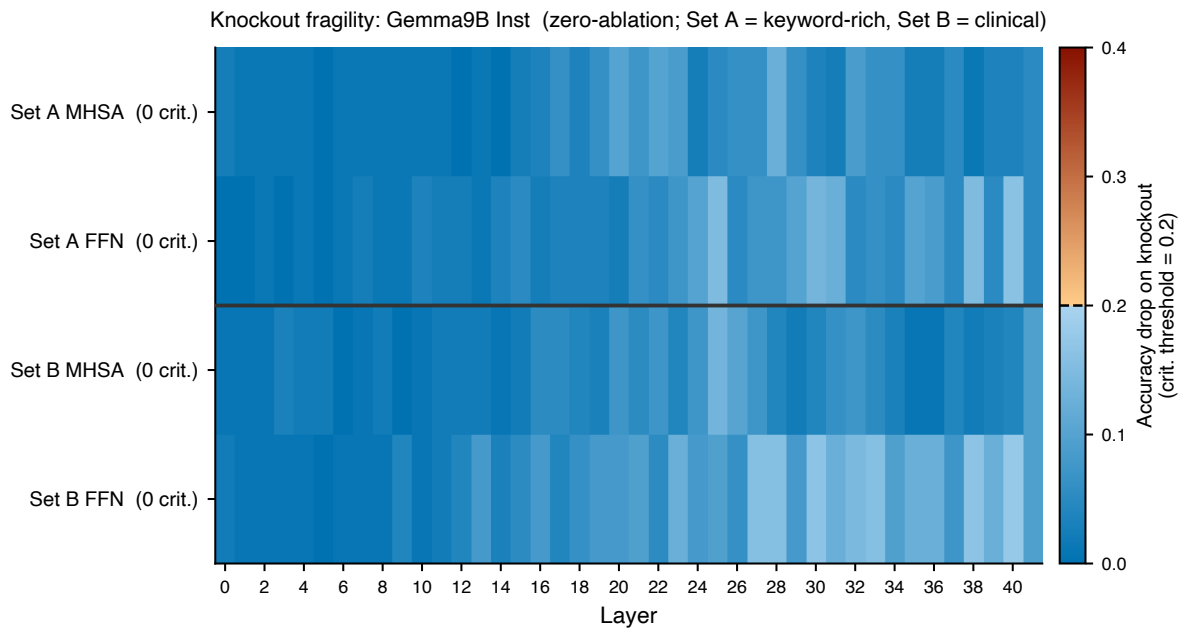

Figure 8: Layer knockout fragility in Gemma-2-9B Instruct (zero-ablation). Each cell shows the accuracy drop caused by zeroing a single layer's output—MHSA (top two rows) or FFN (bottom two rows)—for keyword-rich Set A and keyword-free clinical Set B. Colour scale runs from blue (near-zero drop) to dark red (>40% drop); the critical threshold of >20% drop is marked in the colour bar. Gemma-9B shows zero critical layers in either condition, demonstrating the distributed processing architecture characteristic of larger models. This contrasts sharply with Llama-1B Instruct, where the same analysis identifies 12–16 critical layers for Set B (table 10). At larger scale, emotion processing is architecturally redundant: no single layer failure is catastrophic.

emotion categorization operates through the residual stream rather than through differential attention allocation. The emotional content of the stimulus is integrated into the residual stream during standard forward-pass processing; attention at the extraction position then reads from this accumulated representation against the label space. The mechanism that distinguishes Set A from Set B performance—and that separates binary detection from categorical classification—is encoded in *what* the residual stream contains at each layer, not in *where* the model attends.

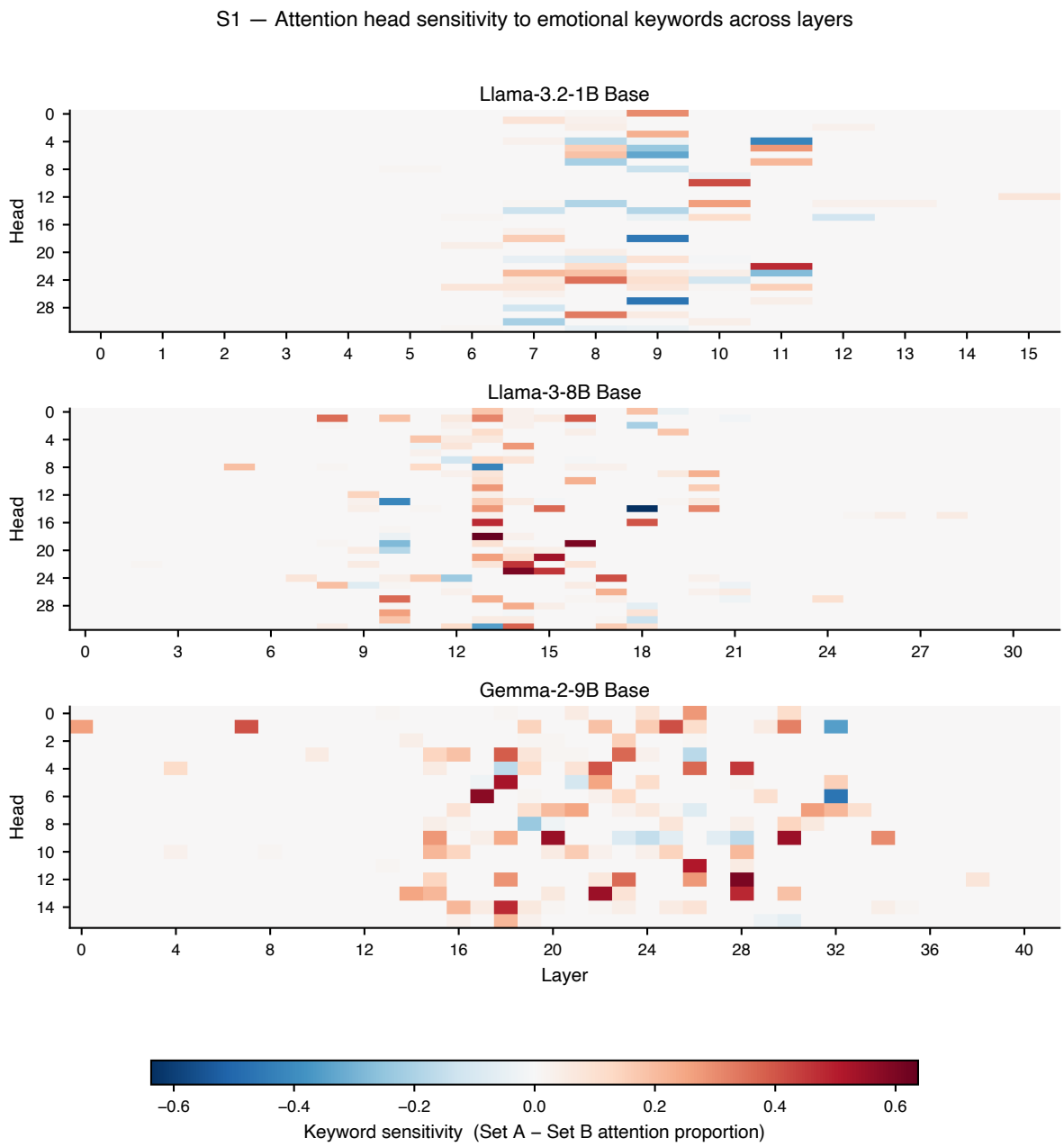

Figure 9: Attention head keyword sensitivity across layers. Each cell shows the difference in attention proportion allocated to emotion keyword tokens between Set A and Set B stimuli (positive = more attention to keywords in Set A; negative = more in Set B). Sensitivity is concentrated in mid-to-late layers—layers 6–12 in Llama-1B, 12–18 in Llama-8B, 16–28 in Gemma-9B. Early layers show near-zero differential sensitivity across both stimulus types, consistent with the model first constructing a general semantic representation before differentiating along keyword presence. The layer-onset of keyword sensitivity corresponds roughly to the affect-reception saturation zone shown in fig. 3.